

Dynamic Host Configuration Protocol (DHCP):

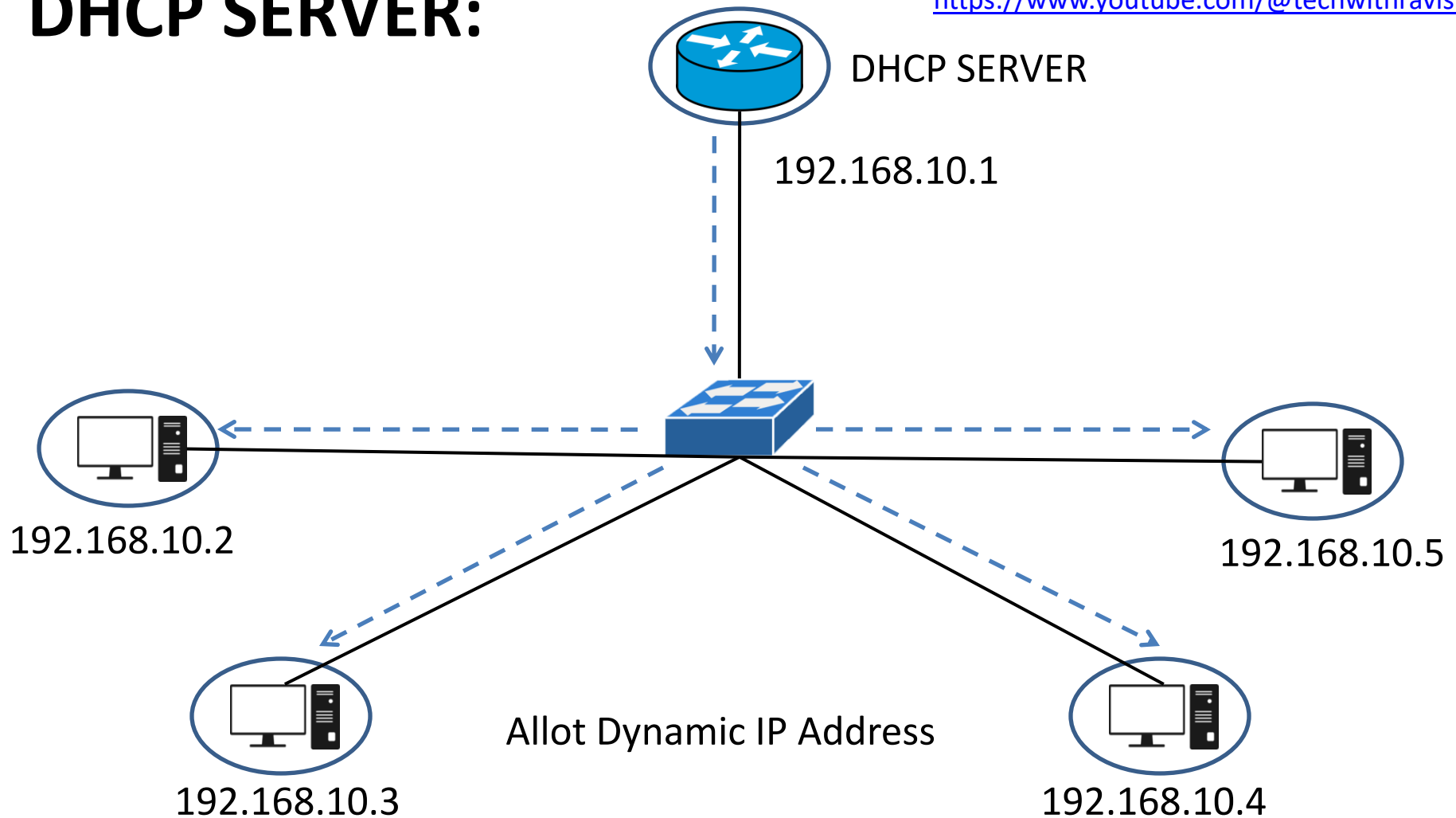
The **Dynamic Host Configuration Protocol (DHCP)** is a network management protocol used on Internet Protocol (IP) networks. DHCP automates the assignment of IP addresses, subnet masks, gateways, and other IP parameters to network devices (clients) from a centralized server (DHCP server).

Key Features of DHCP

- Automatic IP Address Assignment**
- Lease-Based Addressing**
- Centralized Management**
- Additional Configuration Parameters**

DHCP SERVER:

<https://www.youtube.com/@techwithravish>



Benefits of DHCP:

Simplifies Network Management:

- Automates IP address assignment and configuration, reducing administrative overhead.

Reduces Configuration Errors:

- Minimizes manual IP configuration errors and address conflicts.

Efficient IP Address Utilization:

- Dynamically reassigns IP addresses to devices, optimizing address utilization.

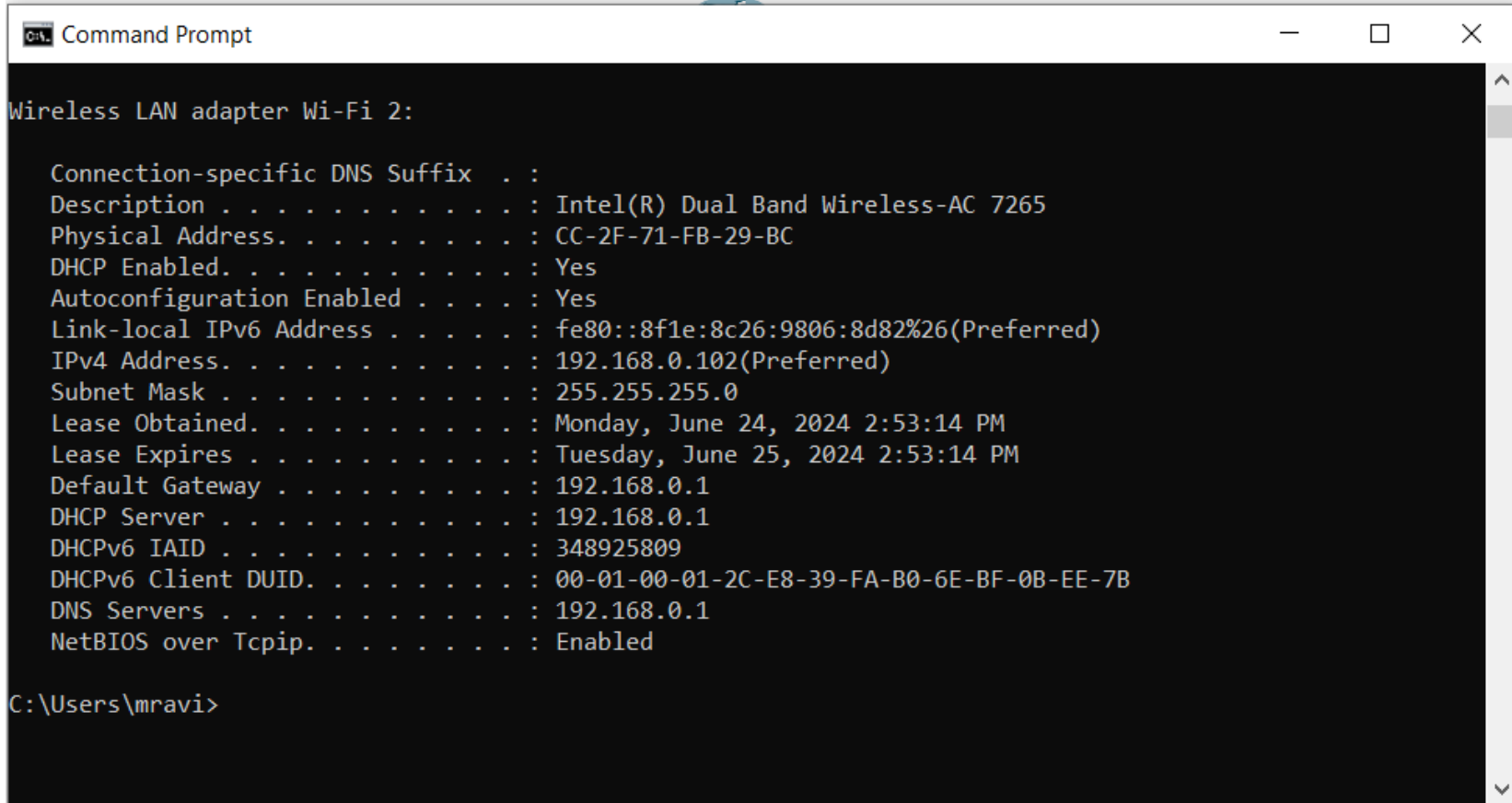
Flexibility:

- Easily scales to accommodate growing networks and changing configurations.

IP Addresses in your PC allotted by DHCP:

C:\users\mravi> ipconfig/all

<https://www.youtube.com/@techwithravish>



```
Command Prompt

Wireless LAN adapter Wi-Fi 2:

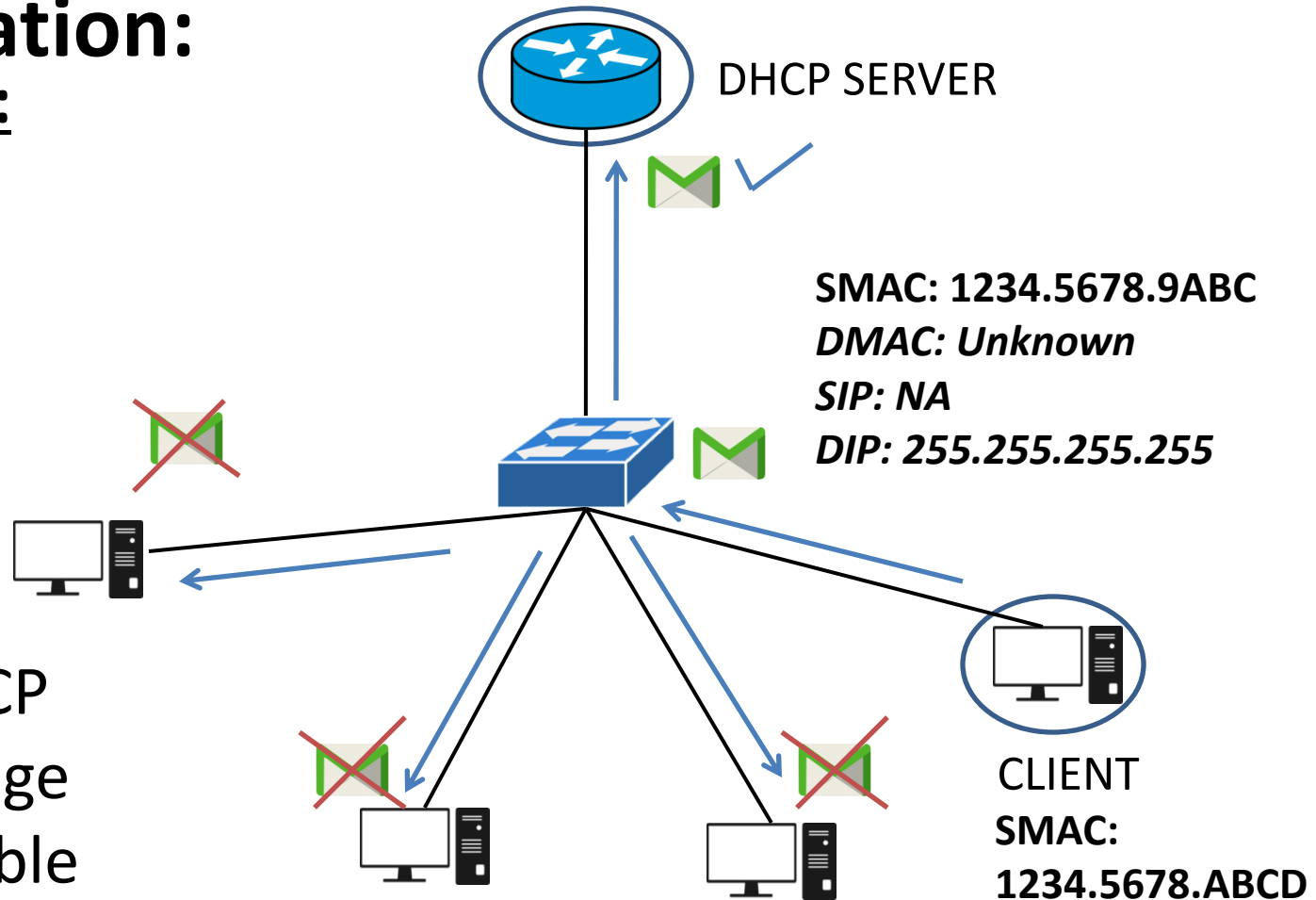
    Connection-specific DNS Suffix  . : 
    Description . . . . . : Intel(R) Dual Band Wireless-AC 7265
    Physical Address. . . . . : CC-2F-71-FB-29-BC
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::8f1e:8c26:9806:8d82%26(Preferred)
    IPv4 Address. . . . . : 192.168.0.102(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Monday, June 24, 2024 2:53:14 PM
    Lease Expires . . . . . : Tuesday, June 25, 2024 2:53:14 PM
    Default Gateway . . . . . : 192.168.0.1
    DHCP Server . . . . . : 192.168.0.1
    DHCPv6 IAID . . . . . : 348925809
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-E8-39-FA-B0-6E-BF-0B-EE-7B
    DNS Servers . . . . . : 192.168.0.1
    NetBIOS over Tcpip. . . . . : Enabled

C:\Users\mravi>
```

DHCP Operation:

DHCP Discover:

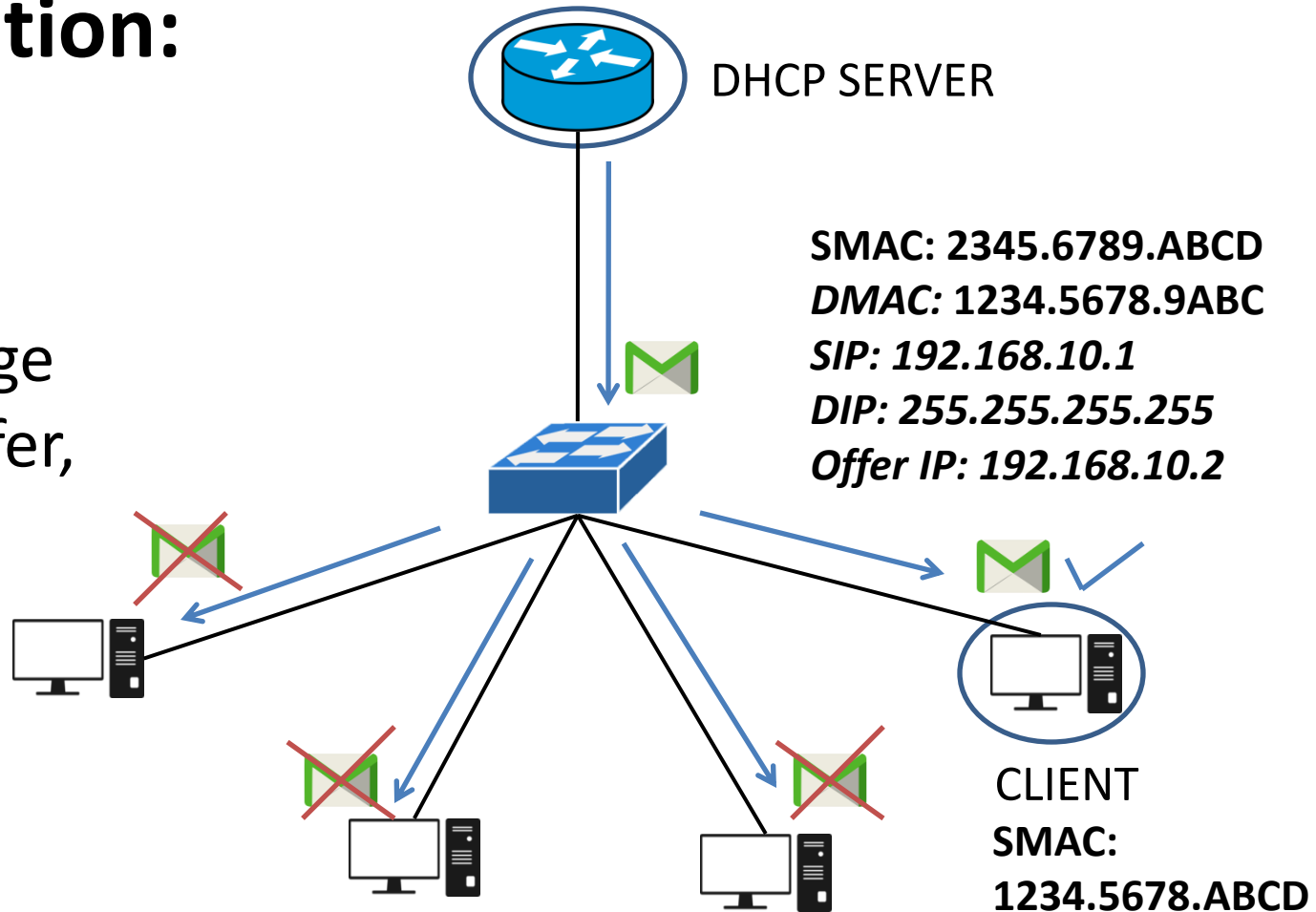
When a client device (e.g., a computer or smartphone) connects to a network, it broadcasts DHCP Discover message to locate available DHCP servers.



DHCP Operation:

DHCP Offer:

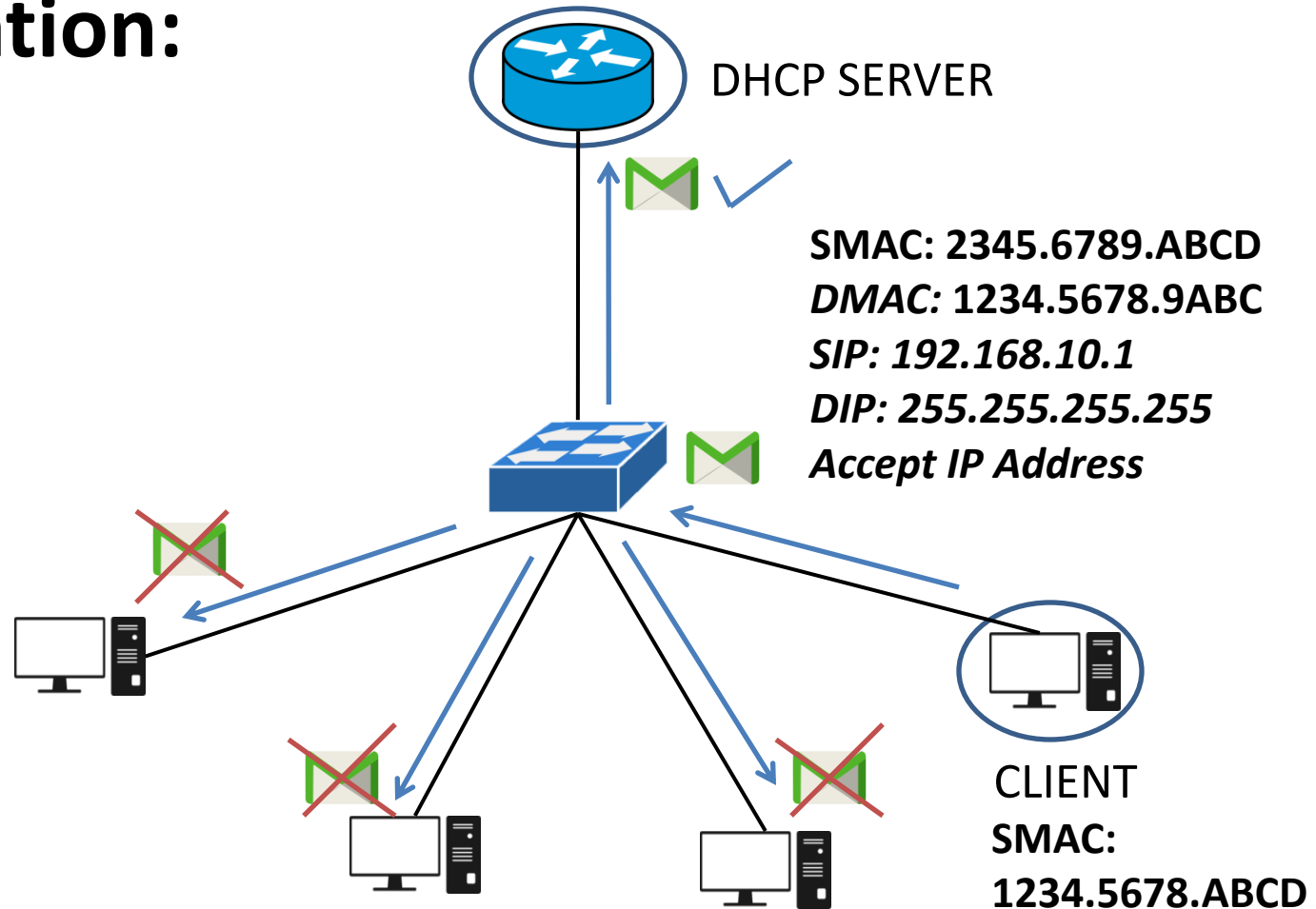
A DHCP server responds to the Discover message with a DHCP Offer, providing an available IP address and additional configuration parameters.



DHCP Operation:

DHCP Request:

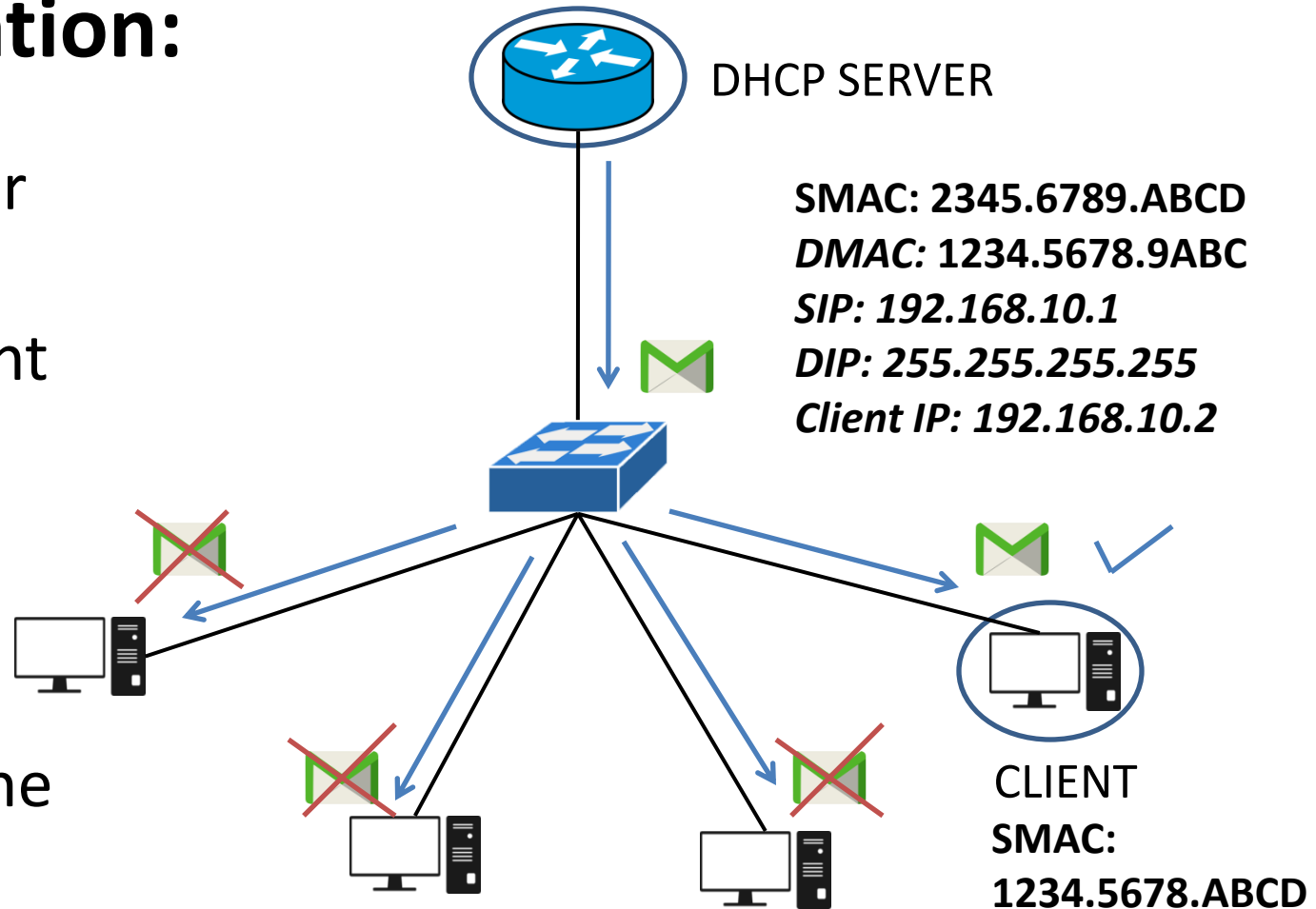
The client responds to the Offer with a DHCP Request message, indicating its acceptance of the offered IP address.



DHCP Operation:

DHCP ACK:

The DHCP server sends a DHCP Acknowledgment message to confirm the assignment of the IP address and complete the configuration process.



DHCP Configuration:

Router> enable

Router# configure terminal

Router(config)# ip dhcp pool MY_POOL

Router(dhcp-config)# network 192.168.1.0 255.255.255.0

Router(dhcp-config)# default-router 192.168.1.1

Router(dhcp-config)# dns-server 8.8.8.8

Router(dhcp-config)# exit

**Router(config)# ip dhcp excluded-address 192.168.1.1
192.168.1.10**